# SARDAR PATEL UNIVERSITY (SPU)

## Department of Computer Science & IT • BCA 4-Year Programme

**SEMESTER 7 • ON-JOB TRAINING (OJT) OFFICIAL STUDENT DAILY LOGBOOK (6 CREDITS)**

### 1. STUDENT TRAINEE IDENTIFICATION

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|---|---|---|---|
| **Candidate Name:** | **[Student Name]** | **SPU Enrollment No:** | **[Enrollment Number]** |
| **Degree Programme:** | **BCA (4-Year Honours)** | **Semester & Credits:** | **Semester 7 • 6.0 Credits** |
| **College / Institute:** | **Affiliated Institute of SPU** | **Faculty Coordinator:** | **[Faculty Guide Name]** |

### 2. HOST ORGANIZATION / COMPANY PROFILE

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|---|---|---|---|
| **Organization Name:** | **[Approved Industry Partner / Company]** | **Industry Domain:** | **Data Analytics & Enterprise IT** |
| **Company Mentor:** | **[Industry Supervisor Name]** | **Supervisor Designation:** | **Senior Data Analyst / Tech Lead** |
| **Training Location:** | **On-Site Development Facility** | **Official SPU Deadline:** | **24 November 2026 (Mandatory)** |

### 3. OJT WORK HOURS & CAPSTONE PROJECT SUMMARY

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|---|---|---|---|
| **Training Duration:** | **12 October 2026 – 24 November 2026** | **Active Working Days:** | **32 Days (Strictly 6.0 Hours / Day)** |
| **Total Certified Hours:** | **180.0 Hours Certified** | **Capstone Project:** | **TELEPULSE 360° Telecom Platform** |
| **Dashboard Stack:** | **Streamlit Cloud + Plotly Web App** | **Report Submission:** | **SPU 10-Section Dossier + Viva** |

**SPU Semester 7 OJT Mandatory Directives:**

- OJT carries **6 credits (180 verified hours)** as a core subject in BCA Semester 7.
- Daily entries must document 6.0 working hours split across theoretical mastery, hands-on lab programming, dashboard making skills, and capstone project development.
- Trainee must submit this certified Logbook along with the official 10-Section Project Report Dossier, Presentation Deck (PPT), and appear for the Final Viva Voce examination before the University Board before **24 November 2026**.

| Trainee Student Signature | Industry Mentor (Seal & Sign) | SPU Faculty Coordinator / HOD |
|---|---|---|

## OFFICIAL DAILY WORK LOG SHEETS (DAY 01 TO DAY 32)

Certified 6.0 Hours / Day • SPU BCA Sem 7 (180 Hours Total)

**Working Day 01** **12 Oct 2026** (Monday) • Cal Day 01 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 6h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Python Data Stack Setup & NumPy Foundations**

**THEORY & CONCEPT (2.0H):** Python data science ecosystem ka overview samjha; VS Code aur Jupyter Notebook kernel configuration seekhi. Python native lists (pointer-based) aur NumPy contiguous memory buffer (C-level execution speed) ke beech performance difference study kiya. Array dimensionality (1D, 2D, 3D), ndarray attributes (shape, size, ndim, dtype) aur numeric memory allocations (int32 vs float64) ka theoretical concept clear kiya.

**HANDS-ON LAB & CODE (4.0H):** Jupyter lab me numpy initialize karke `np.array()`, `np.zeros((3,4))`, `np.ones((2,5))`, `np.arange(10,50,2)`, aur `np.linspace(0,1,100)` execute kiye. Created arrays ke `.shape`, `.dtype`, `.itemsize`, aur `.nbytes` print karke memory consumption audit verify kiya. Day 1 ka environment validation log complete kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 02** **13 Oct 2026** (Tuesday) • Cal Day 02 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 12h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **NumPy Indexing, Slicing & Boolean Masking**

**THEORY & CONCEPT (2.0H):** Multi-dimensional arrays me data extraction aur slicing mechanics ko deeply study kiya. Memory View vs Deep Copy ka concept samjha ki basic slicing view return karti hai jisse original memory link rehti hai. Conditional filtering ke liye Boolean Masking ka mathematical logic samjha, jisme bitwise operators (&, |, ~) ke through complex logic execute hoti hai.

**HANDS-ON LAB & CODE (4.0H):** 5x5 numeric matrix create karke row-column sub-setting ki (`arr[1:4, 2:5]`, step strides `arr[::2, ::-1]`). Condition-based filtering execute ki jaise `arr[arr > 50]` aur multiple boolean criteria evaluate kiye. `np.where()` function use karke conditional value replacement implement kiya. 10 practical matrix extraction problems solve kiye.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 03** **14 Oct 2026** (Wednesday) • Cal Day 03 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 18h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Array Manipulation, Reshaping & Stacking**

**THEORY & CONCEPT (2.0H):** Array structural reorganization aur dimension manipulation algorithms padhe. `.reshape()` execute karne ke rules (size consistency) aur Row-Major (C-style) vs Column-Major (Fortran-style) memory layout samjha. Multidimensional array ko 1D me transform karne ke liye `.flatten()` (deep copy) aur `.ravel()` (memory reference) ke beech architectural difference study kiya.

**HANDS-ON LAB & CODE (4.0H):** 24 elements ke 1D array ko (2,3,4), (4,6), aur (8,3) shape matrices me transform kiya. Multi-feature dataset simulation ke liye `np.vstack()`, `np.hstack()`, `np.dstack()`, aur `np.concatenate()` execute karke vertical aur horizontal array combining ki. `np.split()` aur `np.hsplit()` se arrays ko train-test splits jaisi chunks me divide kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 04** **15 Oct 2026** (Thursday) • Cal Day 04 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 24h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Vectorization, Broadcasting & Mathematical Operations**

**THEORY & CONCEPT (2.0H):** Vectorization aur Broadcasting Rules ka mathematical framework samjha. Traditional Python loops ke comparison me vectorized operations C-loops par fast execute hote hain. Broadcasting ke dimensions rules (trailing axes equality ya dimension 1) samjhe. Universal functions (ufuncs) aur axis-based summary statistics (`axis=0` columns, `axis=1` rows) ka concept clear kiya.

**HANDS-ON LAB & CODE (4.0H):** Incompatible shapes jaise (4,3) matrix aur (1,3) vector par arithmetic broadcasting test kiya. Statistical functions execute kiye: `np.mean()`, `np.median()`, `np.std()`, `np.var()`, aur `np.percentile()` along specific axes. Linear algebra me dot product (`np.dot`), matrix multiplication (`@`), aur transpose (`.T`) verify kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 05** **16 Oct 2026** (Friday) • Cal Day 05 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 30h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Random Sampling & Introduction to Pandas**

**THEORY & CONCEPT (2.0H):** Probabilistic random sampling aur reproducibility ke liye `np.random.seed()` ka role samjha. Core analytics library Pandas ka introduction start hua. Tabular business data store karne ke liye do primary data structures: 1D labeled `Series` aur 2D tabular grid `DataFrame` ki architecture study ki. Explicit Index vs Implicit Range Index mechanics samjhe.

**HANDS-ON LAB & CODE (4.0H):** `np.random.normal()`, `np.random.randint()`, aur `np.random.choice()` use karke 1,000 synthetic transaction records generate kiye aur binary `.npy/.npz` me save/load kiya. In arrays ko Python dictionary me pack karke pehla Pandas DataFrame construct kiya. Index attributes, column headers, aur `.dtypes` inspect kiye.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 06** **17 Oct 2026** (Saturday) • Cal Day 06 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 36h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Pandas Data Ingestion & Structural Inspection**

**THEORY & CONCEPT (2.0H):** External data sources se tabular data ingest karne aur data integrity verify karne ke methods explore kiye. Alag-alag file formats (CSV, Excel `.xlsx`, JSON) ke ingestion parameters jaise `encoding='utf-8'`, `delimiter`, `na_values`, aur `parse_dates` study kiye. Large datasets ke memory footprint ko evaluate karne ki importance samjhi.

**HANDS-ON LAB & CODE (4.0H):** 50,000+ records ka Telecom Transaction CSV dataset Pandas me ingest kiya (`pd.read_csv()`). Structural inspection functions run kiye: `.head(10)`, `.tail(10)`, `.shape`, `.columns`, aur `.info()` jisse null counts aur dtypes verify huye. `.describe()` run karke mean, std dev, min, 25%, 50%, 75%, max values ka summary extract kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 07** **19 Oct 2026** (Monday) • Cal Day 08 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 42h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Data Selection & Querying with .loc and .iloc**

**THEORY & CONCEPT (2.0H):** Pandas me precise row/column indexing aur sub-setting techniques study ki. Do core indexers ka fundamental difference samjha: **`.loc[]`** jo purely label-based hota hai (index name aur column header ke basis par filter karta hai) aur **`.iloc[]`** jo integer position-based (0-indexed position) hota hai. `df.query()` syntax aur boolean mask efficiency compare ki.

**HANDS-ON LAB & CODE (4.0H):** `.loc` ke through specific date ranges aur region columns extract kiye, jabki `.iloc` se top 100 rows aur positional slices extract kiye. Multi-condition queries run ki jaise: amount > 500 aur status 'SUCCESS' (`df[(df['amount'] > 500) & (df['status'] == 'SUCCESS')]`). `.isin()` aur `.str.contains()` filtering use karke 12 practical business scenarios solve kiye.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 08** **21 Oct 2026** (Wednesday) • Cal Day 10 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 48h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Data Cleaning: Missing Values & Duplicates**

**THEORY & CONCEPT (2.0H):** Data quality audit aur cleaning pipeline methodology study ki. Missing data (NaN, None, NaT) ke causes samjhe: MCAR (Missing Completely at Random) vs MNAR (Missing Not at Random). Drop strategy (`dropna()`) kab use karni hai aur Statistical Imputation (`fillna()` with Mean for normal distribution, Median for skewed, Mode for categorical) ka decision framework establish kiya.

**HANDS-ON LAB & CODE (4.0H):** Dataset par `.isna().sum()` aur `.isnull().mean() * 100` run karke missing % report banayi. Critical columns me jahan user ID missing thi drop kiya; delivery/recharge fee me median value impute ki. Duplicate records identify karne ke liye `df.duplicated()` run kiya aur `drop_duplicates(subset=['transaction_id'], keep='first')` execute karke 100% data integrity achieve ki.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 09** **22 Oct 2026** (Thursday) • Cal Day 11 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 54h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Type Casting & Vectorized Text Normalization**

**THEORY & CONCEPT (2.0H):** Memory optimization aur string cleaning techniques study ki. Uncleaned datasets me numeric prices aur dates aksar object (string) format me aate hain jisse computation error hota hai. `astype()` aur `pd.to_numeric(errors='coerce')` ke underlying behavior ko samjha. Vectorized `.str` accessor methods aur regex pattern substitution ke principles padhe.

**HANDS-ON LAB & CODE (4.0H):** Price column se currency symbols (₹, commas) strip karke `float64` me cast kiya. Categorical columns (Operator, Plan Type, Circle) par `.str.strip()`, `.str.lower()`, aur `.str.title()` standardise kiye. Categorical text columns ko `category` dtype me convert karke overall DataFrame memory consumption 40% reduce kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 10** **23 Oct 2026** (Friday) • Cal Day 12 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 60h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Pandas GroupBy & Multi-Level Aggregations**

**THEORY & CONCEPT (2.0H):** Data analytics ka core pattern—**Split-Apply-Combine Paradigm**—deeply study kiya. Dataset ko categorical keys par split karke summary statistics (Sum, Mean, Count, Min, Max) compute karne ka workflow samjha. `.agg()` method ke through multiple columns par simultaneous custom aggregations aur lambda functions apply karne ke rules clear kiye.

**HANDS-ON LAB & CODE (4.0H):** Operator-wise total sales: `df.groupby('operator')['amount'].sum()` compute kiya. Multi-level grouping execute ki: `df.groupby(['operator', 'circle']).agg({'amount': ['sum', 'mean'], 'transaction_id': 'count'})`. Hierarchical column names ko flatten kiya aur top-performing telecom circles aur plans ka ranking table prepare kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 11** **24 Oct 2026** (Saturday) • Cal Day 13 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 66h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Relational Data Merging, Joins & Concatenation**

**THEORY & CONCEPT (2.0H):** Relational database joins ko Pandas me apply karne ke methods padhe. Multi-table enterprise architecture (Customers, Plans, Transactions) me Joins ki zaroorat samjhi. 4 join types: **Inner Join**, **Left Outer Join**, **Right Outer Join**, aur **Full Outer Join** ke mathematical Venn-diagrams study kiye. `pd.concat()` ke axis=0 aur axis=1 mechanics samjhe.

**HANDS-ON LAB & CODE (4.0H):** 3 alag-alag datasets load kiye: Customers (50k rows), Transactions, aur Plans Master. `pd.merge(transactions, customers, on='customer_id', how='left')` se customer demographics merge kiye; fir plan specifications ko `plan_id` par join kiya. Suffix overlaps (_x, _y) clean karke Master Analysis Table banaya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 12** **26 Oct 2026** (Monday) • Cal Day 15 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 72h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Pivot Tables & Wide-to-Long Reshaping (Melt)**

**THEORY & CONCEPT (2.0H):** Multi-dimensional business reporting ke liye **Pivot Tables** aur reshaping operations explore kiye. `pd.pivot_table()` me multi-index rows, columns, aur aggregation margins (Grand Totals) design karna samjha. Wide format data (months as columns) ko unpivot karke analysis-friendly long format me convert karne ke liye `pd.melt()` ka algorithmic mechanism study kiya.

**HANDS-ON LAB & CODE (4.0H):** `pd.pivot_table(df, values='amount', index='operator', columns='payment_mode', aggfunc='sum', fill_value=0, margins=True)` run karke Operator vs Payment Mode ka Cross-Tabular Matrix banaya. Wide financial summary ko `pd.melt(id_vars=['operator'], value_vars=['Oct_Revenue', 'Nov_Revenue'])` se long format me transform kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 13** **27 Oct 2026** (Tuesday) • Cal Day 16 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 78h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **DateTime Parsing & Temporal Time-Series Analysis**

**THEORY & CONCEPT (2.0H):** Temporal data analytics aur time-series handling ke concepts padhe. Timestamps ko string ki jagah `datetime64[ns]` me parse karne ki importance samjhi taaki date arithmetic aur lag calculations possible hon. Date components extraction (`.dt.day_name()`, `.dt.hour`) aur business periodicity ke according frequency resampling (`resample('D')`, `resample('W')`) study kiye.

**HANDS-ON LAB & CODE (4.0H):** `pd.to_datetime(df['recharge_time'])` execute karke transaction timestamps parse kiye. Feature engineering columns banaye: `Hour_of_Day`, `Day_of_Week`, aur `Is_Weekend`. Recharge date index par set karke daily revenue velocity calculate ki. Paya gaya ki month-ends aur weekends par recharge activity me 35% spike aata hai.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 14** **28 Oct 2026** (Wednesday) • Cal Day 17 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 84h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Capstone Project: Telecom Dataset & Synopsis Design**

**THEORY & CONCEPT (2.0H):** College capstone project ke formal scope, problem statement, aur business objectives ko finalize kiya. Project title select kiya: **'TELEPULSE 360°: Pan-India Telecom Intelligence, ARPU Optimization & Consumer Churn Platform'**. 5 core research questions formulate kiye (Recharge cycle decay, payment gateway drop-offs, ARPU across Jio/Airtel/Vi/BSNL, 4G/5G booster trends).

**HANDS-ON LAB & CODE (4.0H):** 50,000+ transaction records ka relational schema validate kiya. Har variable ki definition, data type, allowed range, aur foreign key constraints document karke formal **Data Dictionary** banayi. College presentation aur faculty evaluation ke liye 2-page ka official **Project Synopsis** draft kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 15** **29 Oct 2026** (Thursday) • Cal Day 18 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 90h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Outlier Analysis (IQR / Z-Score) & Cleaning Pipeline**

**THEORY & CONCEPT (2.0H):** Statistical outlier detection techniques study ki jo data distribution ko clean karne me use hoti hain. **Z-Score** method (standard normal distribution, threshold $\pm 3\sigma$) aur **IQR (Interquartile Range) Method** ($Q_1 - 1.5 \times IQR$ to $Q_3 + 1.5 \times IQR$, skewed business data ke liye robust) ka mathematical comparison kiya.

**HANDS-ON LAB & CODE (4.0H):** Numeric columns ('amount', 'delay_days', 'data_consumed') par IQR boundaries calculate kiye. Identified extreme values ko verify kiya ki wo genuine high-value annual recharges (₹2999) hain ya system errors. End me ek modular Python function `clean_dataset_pipeline(raw_df)` likha jo raw data input lekar fully cleaned output return karta hai.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 16** **30 Oct 2026** (Friday) • Cal Day 19 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 96h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Data Visualization Architecture with Matplotlib**

**THEORY & CONCEPT (2.0H):** Python foundational plotting library **Matplotlib** aur uske **Object-Oriented Architecture** ko study kiya. `pyplot` state-machine interface vs `Figure` & `Axes` OOP approach ke beech ka farak samjhaya. Visual hierarchy, color theory, labels, tick locators, legends, aspect ratio, aur chart integrity ke principles samjhe.

**HANDS-ON LAB & CODE (4.0H):** `fig, ax = plt.subplots(figsize=(10,6))` structure use karke Line Plots (Monthly revenue trend), Bar Charts (Operator sales), aur Histograms (Recharge amount distribution) banaye. Customizations implement kiye: `ax.grid(True)`, custom palettes, aur bar-top annotations. Charts ko publication-quality resolution (`dpi=300`) me export kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 17** **02 Nov 2026** (Monday) • Cal Day 22 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 102h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Statistical Visualizations with Seaborn**

**THEORY & CONCEPT (2.0H):** Matplotlib ke upar bani high-level statistical visualization library **Seaborn** ko study kiya. Seaborn direct Pandas DataFrames ke sath integrate hoti hai aur automatically statistical aggregations, confidence intervals, aur semantic mappings (hue, size) handle karti hai. Distribution plots aur Categorical comparison plots ke statistical significance ko samjha.

**HANDS-ON LAB & CODE (4.0H):** `sns.set_theme(style='whitegrid')` apply kiya. Alag-alag operators ke recharge ticket size compare karne ke liye Boxplots banaye (`sns.boxplot(x='operator', y='amount', data=df)`) jisse median aur quartiles clear dikhe. Recharge frequency density ke liye KDE plot aur payment modes distribution ke liye `sns.countplot` visualize kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 18** **03 Nov 2026** (Tuesday) • Cal Day 23 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 108h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Multivariate Analysis, Covariance & Heatmaps**

**THEORY & CONCEPT (2.0H):** Multiple numerical variables ke aapsi correlation ko quantify karne ke liye **Pearson Correlation Coefficient ($r$)** ka concept samjha (range -1 to +1). Correlation vs Causation ka difference study kiya. Multicollinearity kya hoti hai aur correlation matrix ke visual representation ke liye annotated Heatmaps aur Facet Grids ka use kaise hota hai, ispar in-depth study ki.

**HANDS-ON LAB & CODE (4.0H):** Numerical features ka correlation matrix calculate kiya (`corr_matrix = df[['amount', 'validity', 'daily_data', 'recharge_gap']].corr()`). Seaborn se stunning diverging heatmap plot kiya (`sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')`). Paya gaya ki validity aur amount me high correlation ($r=0.88$) hai. Insights document kiye.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 19** **04 Nov 2026** (Wednesday) • Cal Day 24 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 114h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Comprehensive Exploratory Data Analysis (EDA) Notebook**

**THEORY & CONCEPT (2.0H):** Pure EDA methodology ko end-to-end structure kiya: **Univariate Analysis** (single variable characteristics), **Bivariate Analysis** (pairwise relationships), aur **Multivariate Analysis** (multi-dimensional interactions). Har chart ke neeche formal business interpretation aur executive observations likhne ke guidelines samjhe.

**HANDS-ON LAB & CODE (4.0H):** Professional Jupyter Notebook (`01_telecom_eda.ipynb`) me 50,000 records par comprehensive EDA run kiya. Top findings: (1) 68% recharges UPI se huye, (2) 28-day plan users 3-4 din ka gap lete hain renewal me, (3) Jio me micro data boosters volume highest hai, (4) Airtel ka ARPU highest (₹214) hai. Markdown commentary compile ki.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 20** **05 Nov 2026** (Thursday) • Cal Day 25 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 120h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **SQL Foundations for Data Analysts & SQLite Modeling**

**THEORY & CONCEPT (2.0H):** Data Analysis ke core skill—**Structured Query Language (SQL)**—ki shuruat ki. Relational Database Management Systems (RDBMS), ACID properties, Table Schemas, Primary Keys, Foreign Keys, aur Entity-Relationship (ER) modeling study ki. Enterprise production me CSVs ki jagah relational SQL databases kyu use hote hain, iski theoretical importance samjhi.

**HANDS-ON LAB & CODE (4.0H):** Python ke `sqlite3` aur `SQLAlchemy` ka use karke local database `telecom_warehouse.db` create kiya. Cleaned DataFrames ko relational tables (`customers`, `plans`, `recharge_transactions`) me dump kiya. Basic DDL/DQL queries run ki: `SELECT * FROM recharge_transactions LIMIT 10` aur `SELECT DISTINCT operator FROM customers`.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 21** **06 Nov 2026** (Friday) • Cal Day 26 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 126h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **SQL Filtering, Sorting & Pagination (Dhanteras Working Day)**

**THEORY & CONCEPT (2.0H):** SQL me conditional query filtering aur result ordering ke mechanics deeply practice kiye. `WHERE` clause me logical operators (`AND`, `OR`, `NOT`), range evaluation (`BETWEEN`), set membership (`IN`), aur text wildcards (`LIKE '%4G%'`) ke rules samjhe. Pagination ke liye `ORDER BY`, `LIMIT`, aur `OFFSET` clauses ka use padha.

**HANDS-ON LAB & CODE (4.0H):** Targeted business queries execute ki: UPI payment mode se kiye gaye ₹500 se bade successful transactions retrieve kiye; Jio aur Airtel ke 84-day plans filter kiye; transactions ko amount ke descending order me sort karke top 20 highest-paying consumers extract kiye (`ORDER BY amount DESC LIMIT 20`). Results ko Pandas DataFrame me fetch kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 22** **11 Nov 2026** (Wednesday) • Cal Day 31 | CORE ANALYTICS & LAB | **6.0 Hours (Cumulative: 132h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **SQL Aggregations & GROUP BY / HAVING (Post-Diwali Session)**

**THEORY & CONCEPT (2.0H):** Post-Diwali session me SQL aggregate functions aur grouping mechanics deeply study kiye. `COUNT()`, `SUM()`, `AVG()`, `MIN()`, `MAX()` ke handling rules dekhe. **`WHERE` vs `HAVING` clause ka difference**: `WHERE` individual rows par filter lagata hai grouping se pehle, jabki `HAVING` aggregated summary values par filter lagata hai grouping ke baad.

**HANDS-ON LAB & CODE (4.0H):** Complex enterprise analytical query likhi: `SELECT circle, operator, COUNT(transaction_id) AS total_recharges, ROUND(AVG(amount), 2) AS arpu, SUM(amount) AS total_revenue FROM recharge_transactions WHERE status = 'SUCCESS' GROUP BY circle, operator HAVING total_revenue > 100000 ORDER BY total_revenue DESC;`. Circle-wise revenue distribution analyze kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 23** **12 Nov 2026** (Thursday) • Cal Day 32 | DASHBOARD SKILL + PROJECT | **6.0 Hours (Cumulative: 138h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Dashboard Making Skill: Design Systems & Multi-Table Relational Project**

**DASHBOARD MAKING SKILL (2.0H):** Modern Business Intelligence & Web Dashboard Design Systems study kiye. Information architecture, F-pattern visual reading flow, 60-30-10 color rule for financial metrics, and typography hierarchy design kiya. Streamlit UI layout principles (`st.set_page_config(layout='wide')`, responsive containers) ka foundation establish kiya.

**CAPSTONE PROJECT (4.0H):** 50,000+ Telecom records ka 3-Table Relational Join execute kiya: `SELECT c.customer_id, c.operator, c.circle, p.plan_name, t.amount, t.payment_mode FROM recharge_transactions t INNER JOIN customers c ON t.customer_id = c.customer_id INNER JOIN plans p ON t.plan_id = p.plan_id WHERE t.status = 'SUCCESS';`. Multi-table foreign keys validate kiye.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 24** **13 Nov 2026** (Friday) • Cal Day 33 | DASHBOARD SKILL + PROJECT | **6.0 Hours (Cumulative: 144h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Dashboard Making Skill: Reactive Filtering & Advanced SQL Window Functions**

**DASHBOARD MAKING SKILL (2.0H):** Dashboard me Reactive State Management aur Dynamic Sidebar Controls create karna seekha. `st.sidebar.selectbox` (All Operators vs Jio vs Airtel vs Vi vs BSNL), dynamic multi-select circle filters, aur date-range pickers ka live binding code likha jo bina page refresh ke instant reactive data slices return karta hai.

**CAPSTONE PROJECT (4.0H):** Customer churn aur recharge delay gap calculate karne ke liye `LAG()` window function use kiya: `LAG(recharge_time) OVER (PARTITION BY customer_id ORDER BY recharge_time)`. Plan expiry ke baad renewal gap > 7 days wale 3,840 high-risk churn subscribers identify kiye.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 25** **14 Nov 2026** (Saturday) • Cal Day 34 | DASHBOARD SKILL + PROJECT | **6.0 Hours (Cumulative: 150h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Dashboard Making Skill: Caching & High-Performance Data Pipeline**

**DASHBOARD MAKING SKILL (2.0H):** Big Data Web Dashboards me latency optimization seekha. Streamlit ka memory caching decorator `@st.cache_data(ttl=3600)` implement kiya taaki 50,000+ rows baar-baar disk se read na hon. CSV vs Parquet format benchmark kiya (Parquet ingestion 6.8x faster aur memory footprint 55% lower paya gaya).

**CAPSTONE PROJECT (4.0H):** Automated Python ETL Pipeline (`etl_pipeline.py`) develop kiya jo raw telecom transaction logs ko parse, clean, type-cast karke `telecom_warehouse.db` SQLite warehouse me auto-load karta hai. Data quality validation test cases pass kiye.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 26** **16 Nov 2026** (Monday) • Cal Day 36 | DASHBOARD SKILL + PROJECT | **6.0 Hours (Cumulative: 156h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Dashboard Making Skill: Executive KPI Metric Cards & ARPU Engine**

**DASHBOARD MAKING SKILL (2.0H):** Executive C-Suite KPI Metric Cards design karne ka skill seekha: `st.metric(label, value, delta)`. Custom CSS styling apply karke glassmorphism KPI tiles banaye jisme Gross Merchandise Value (GMV), Industry ARPU, Total Active Subscribers, aur Transaction Success Rate % positive/negative trend arrows ke sath render huye.

**CAPSTONE PROJECT (4.0H):** Operator-wise competitive benchmarking SQL analytics code compile kiya: Airtel (ARPU: ₹214), Jio (ARPU: ₹188, Volume Leader: 44%), Vi (ARPU: ₹142, Churn Risk: 18%), aur BSNL (ARPU: ₹104). MoM revenue growth rate matrices export kiye.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 27** **17 Nov 2026** (Tuesday) • Cal Day 37 | DASHBOARD SKILL + PROJECT | **6.0 Hours (Cumulative: 162h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Dashboard Making Skill: Interactive Plotly Visualizations & Funnel Diagnostics**

**DASHBOARD MAKING SKILL (2.0H):** Advanced interactive data visualization engineering with Plotly Express & Graph Objects. Hover data formatting (custom tooltip templates), dual-axis plotting (Revenue Bars + Margin Line), dynamic chart zooming, aur color gradient scaling ke production patterns master kiye.

**CAPSTONE PROJECT (4.0H):** Fintech Payment Gateway reliability diagnostics complete kiye: UPI vs Debit Cards vs NetBanking vs Wallets ka Interactive Drop-off Funnel Chart (Initiated -> Gateway Handoff -> Bank Auth -> Success) banaya. Peak-hour server timeout spike (8:30 PM to 10:30 PM) quantify kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 28** **18 Nov 2026** (Wednesday) • Cal Day 38 | DASHBOARD SKILL + PROJECT | **6.0 Hours (Cumulative: 168h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Dashboard Making Skill: Geospatial Mapping & RFM-D ML Clustering**

**DASHBOARD MAKING SKILL (2.0H):** Geospatial Telecom Circle Mapping skill seekha: Plotly Choropleth Map of India integrate kiya. State/Circle boundary GeoJSON data bind karke color intensity ke basis par recharge density aur operator market share show kiya. Hover karne par state-specific telecom metrics render huye.

**CAPSTONE PROJECT (4.0H):** Telecom RFM-D (Recency, Frequency, Monetary + Daily Data Intensity) segmentation model code kiya. Scikit-learn se K-Means Clustering ($k=4$) train kiya with `StandardScaler`, Elbow Curve, aur Silhouette Score evaluation. 3D Plotly cluster scatter view prepare kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 29** **19 Nov 2026** (Thursday) • Cal Day 39 | DASHBOARD SKILL + PROJECT | **6.0 Hours (Cumulative: 174h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Dashboard Making Skill: Multi-Page App Assembly & Navigation**

**DASHBOARD MAKING SKILL (2.0H):** Production Streamlit Multi-Page Directory Architecture build ki: `pages/1_` 📈 `_Executive_Overview.py`, `pages/2_` 🗺️ `_Circle_Geospatial.py`, `pages/3_` 👥 `_Customer_Segmentation.py`, aur `pages/4_` 🎯 `_Smart_Simulator.py`. Clean navigation bar aur consistent branding apply kiye.

**CAPSTONE PROJECT (4.0H):** Main application `app.py` assemble ki. Sabhi analytical pipelines, SQL queries, ML models aur Plotly charts ko real-time reactive filters ke sath connect karke end-to-end local testing perform ki. Responsive testing verify ki.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 30** **20 Nov 2026** (Friday) • Cal Day 40 | 180H TARGET MET! | **6.0 Hours (Cumulative: 180h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Dashboard Making Skill: What-If Simulators & 180h Certified Milestone**

**DASHBOARD MAKING SKILL (2.0H):** Interactive What-If Decision Simulators design karne ka skill seekha: `st.slider` aur mathematical sensitivity models connect karke dynamic prediction widgets banaye. User input change hote hi real-time financial ROI recalculate karne ka reactive engine banaya.

**CAPSTONE PROJECT (4.0H - 180H TARGET MET!):** Do interactive decision engines deploy kiye: (1) **Smart Plan Recommender** jo consumer usage ke hisaab se optimal cost-saving plan recommend karta hai; (2) **Tariff Hike Simulator** jo 10%-25% tariff increase par churn loss vs net profit calculate karta hai. **Mandatory 180.0 Certified OJT Hours achieve huye!**

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 31** **21 Nov 2026** (Saturday) • Cal Day 41 | DASHBOARD SKILL + PROJECT | **6.0 Hours (Cumulative: 180h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **Dashboard Making Skill: Streamlit Cloud Deployment & GitHub Portfolio**

**DASHBOARD MAKING SKILL (2.0H):** Cloud Web Application Deployment & DevOps workflow seekha. `requirements.txt` version lock, `.streamlit/config.toml` production theme settings, aur **Streamlit Community Cloud** par 1-click free cloud hosting execute karke public live portfolio URL generate kiya.

**CAPSTONE PROJECT (4.0H):** Public GitHub repository structure build kiya: modular code (`src/`), Jupyter notebooks, sample data, architecture diagrams, live web app badges, aur comprehensive `README.md`. SPU Presentation Deck (12-slide PPT) finalize kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

**Working Day 32** **23 Nov 2026** (Monday) • Cal Day 43 | DASHBOARD SKILL + PROJECT | **6.0 Hours (Cumulative: 180h)**

**Timing:** 10:00 AM – 01:00 PM & 01:30 PM – 04:30 PM (6.0 Work Hours) **Mode:** On-Site / Lab Mentorship **Supervisor Verification:** Verified

MODULE / TOPIC: **SPU Official Report Dossier (10 Standard Sections) & Faculty Sign-Off**

**DASHBOARD MAKING SKILL (2.0H):** Executive Technical Presentation & Viva Voce Defense skills: Dashboard walkthrough rehearsal, handling technical questions (Streamlit caching, SQL window functions, K-Means mathematical convergence, OLTP vs OLAP architecture), aur faculty evaluation rubrics practice ki.

**CAPSTONE PROJECT (4.0H):** SPU circular format ke according formal **35+ Page OJT Project Report** complete kiya (10 Standard Sections: Coverpage, Certificates, Acknowledgement, Index, Introduction, Objectives, Learning Experience, Outcomes, Conclusions, Attachments). Company Mentor & Faculty sign-off ready kiya.

Trainee Student Signature | Mentor Remarks: **[ Satisfactory / Verified ]** | Company Supervisor Seal & Sign

## CUMULATIVE ATTENDANCE & MENTOR EVALUATION SUMMARY

| WEEK NO. | DATE PERIOD (2026) | CURRICULUM FOCUS & MILESTONES | DAYS LOGGED | HOURS | CUMULATIVE | MENTOR STATUS |
|---|---|---|---|---|---|---|
| **Week 1** | 12 Oct – 17 Oct 2026 | NumPy Architecture, Indexing, Vectorization & Intro to Pandas | 6 Days | 36.0 hrs | 36.0 hrs | **Verified • 100%** |
| **Week 2** | 19 Oct – 24 Oct 2026 | Pandas Selection, Cleaning, Type Casting & Multi-Table Joins (Dussehra 20 Oct Off) | 5 Days | 30.0 hrs | 66.0 hrs | **Verified • 100%** |
| **Week 3** | 26 Oct – 30 Oct 2026 | Pivot Tables, Time-Series, Project Synopsis & Matplotlib (Exam 31 Oct Off) | 5 Days | 30.0 hrs | 96.0 hrs | **Verified • 100%** |
| **Week 4** | 02 Nov – 06 Nov 2026 | Seaborn Statistical Plots, EDA Notebook & SQL Query Foundations | 5 Days | 30.0 hrs | 126.0 hrs | **Verified • 100%** |
| **Week 5** | 11 Nov – 14 Nov 2026 | SQL Aggregations, Joins + **Last 10 Days Dashboard Making & Project Start** | 4 Days | 24.0 hrs | 150.0 hrs | **Verified • 100%** |
| **Week 6** | 16 Nov – 21 Nov 2026 | Streamlit App Architecture, Plotly Funnel, ML Clustering + **180.0 Hours Met!** | 6 Days | 36.0 hrs | **180.0 hrs** | **Target Achieved!** |
| **Week 7** | 23 Nov 2026 | SPU 10-Section Report Dossier, Live Streamlit Cloud Demo & Viva Voce Defense | 1 Day | 6.0 hrs | 180.0+ hrs | **Dossier Ready** |

### INDUSTRY MENTOR FINAL ASSESSMENT RUBRIC

1. **Technical Aptitude (Python & SQL):** Excellent (Grade A+)
2. **Dashboard & Visualization Skills:** Excellent (Grade A+)
3. **Capstone Project Quality:** Outstanding (Enterprise Scale)
4. **Punctuality, Attendance & Logbook:** 100% Attendance Verified
5. **Final OJT Recommendation:** Certified for 6 Credits Award

### SPU 10-SECTION REPORT DELIVERABLES CHECKLIST

✓ Section 1 to 4: Coverpage, Certificates, Acknowledgement, Index
✓ Section 5 & 6: Introduction & 180h Academic Objectives
✓ Section 7: Learning Experience (Theory + Lab + Dashboarding)
✓ Section 8: Learning Outcomes (TELEPULSE 360° Live App)
✓ Section 9: Conclusions, ARPU Findings & Business ROI
✓ Section 10: Attachments (Schema, Queries, Screenshots, PPT)

**Student Trainee Signature** | **Industry Supervisor Seal & Sign** | **SPU Faculty Coordinator / HOD Sign**